



CONTROLLED DRAFT

P2P Protocol & Data Model Specification

Session lifecycle, messages, transfer manifests, local entities and compatibility

Product	Nod
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Release note: This pack is a production-ready baseline. Bracketed operator details must be completed before public release. Legal documents should be reviewed by Nigerian counsel before publication or execution.

1. Protocol objectives

- Human-verifiable peer authentication.
- Explicit content consent after connection consent.
- Versioned forward-compatible control messages.
- Unambiguous payload/manifest correlation.
- Integrity verification and safe failure.
- No dependence on transport delivery ordering between control/file payload classes.

2. Session state machine

IDLE -> READY -> DISCOVERING -> PEER_FOUND -> CONNECTION_REQUESTED -> VERIFYING -> CONNECTED -> CONTENT_PROPOSED -> AWAITING_ACCEPTANCE -> TRANSFERRING -> VERIFYING_PAYLOAD -> COMPLETED -> DISCONNECTED.

Failure states include PERMISSION_DENIED, PEER_LOST, USER_REJECTED, AUTHENTICATION_FAILED, TRANSFER_INTERRUPTED, INSUFFICIENT_STORAGE, INTEGRITY_FAILED, TIMEOUT, CANCELLED and UNSUPPORTED_VERSION.

3. Envelope

Field	Type	Purpose
protocol_version	uint32	Compatibility negotiation.
session_id	UUID/string	Logical connection.
message_id	UUID/string	Deduplication/correlation.
timestamp	int64	Diagnostic ordering only; not trust/security authority.
message_type	enum	HELLO, VERIFY, PROFILE_OFFER, MANIFEST, ACCEPT, REJECT, CANCEL, COMPLETE, ERROR.
body	typed message	Message-specific payload.

4. Transfer manifest

Field	Description
transfer_id	Unique identifier.
items[]	Item IDs with display name, MIME, byte length and SHA-256.
total_bytes	Aggregate bytes for capacity/progress.
content_category	contacts/photos/videos/documents/mixed.
sender_alias	Nearby alias only; full profile shared separately by consent.

5. Transfer rules

1. Sender establishes verified session.
2. Sender sends metadata offer/manifest before large file bytes.
3. Receiver reviews source alias, item count, types and total size.
4. Receiver accepts or rejects.
5. On acceptance, sender begins payload transfer and maps transport payload IDs to manifest item IDs.
6. Receiver writes to private temporary destination while computing/verifying hashes.
7. Only verified files are atomically finalized to MediaStore/Downloads or selected destination.
8. Both parties receive completion/error control messages; temporary files are removed on terminal failures.

6. Profile data model

Entity	Core fields
NodProfile	profileId, displayName, avatarUri, phones[], emails[], company?, jobTitle?, website?, socialLinks[], createdAt, updatedAt
ShareProfile	shareProfileId, label, ordered field-selection rules, isDefault
NearbyIdentity	ephemeralPeerId, nearbyAlias, capability flags; no full contact fields
TransferRecord	transferId, direction, category, itemCount, peerAlias, status, startedAt, completedAt, byteCount
BlockedPeer	stable local fingerprint/identifier where available, alias snapshot, blockedAt

7. Compatibility

- Peers exchange supported protocol version and feature capability flags during HELLO.
- Unsupported major versions fail gracefully before content transfer.
- Optional fields are ignored when unknown unless marked required by negotiated feature.
- No UI should assume transport callback order equals protocol order.